

Assessing Trust in Active Learning Systems: Insights from Query Policies and Uncertainty Visualization

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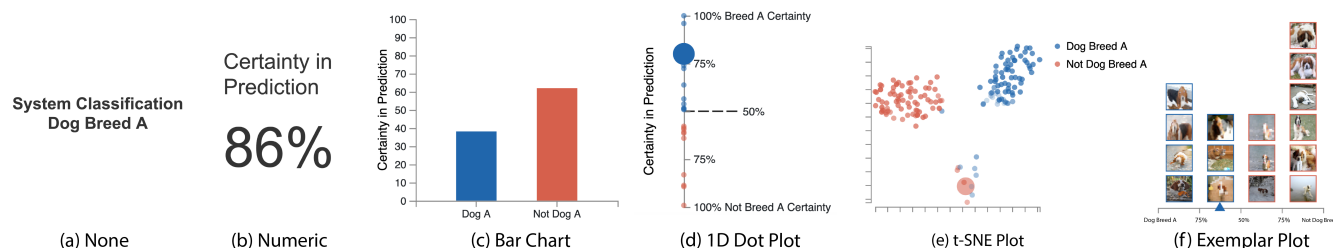


Figure 1: We evaluate the role of classification uncertainty on analyst trust in active learning systems. We visualize classification uncertainties using six different techniques (a)-(f) designed to reveal increasing amounts of information about an algorithm's classification decisions and pair these visualizations with five different query policies.

ABSTRACT

Active learning (AL) systems have become increasingly popular for various applications in machine learning (ML), including medical imaging, environmental monitoring, and geospatial analysis. These systems rely on inputs dynamically queried from people to enhance classification. Ensuring appropriate analyst trust in these systems presents a significant obstacle as analyst over- or underreliance may adversely affect a given application. Common AL strategies enhance classification models by asking analysts to provide labels for data points with the highest degree of uncertainty. However, such model-centric policies do not consider potential priming effects on the analyst and how they will affect people's trust in the system post-training. We present an empirical study assessing how AL query policies and visualizations that enhance transparency in a classifier's decisions influence trust in automated image classifiers. We found that query policy may significantly influence an analyst's perception of system capabilities, while the level of visual transparency into classifier certainty may influence an analyst's ability to perform a classification task. Our study informs the design of interactive labeling systems to help mitigate the effects of overreliance and calibrate appropriate trust in automated systems.

CCS CONCEPTS

• **Human-Centered computing** → **Empirical Studies in HCI**; • **Computing Methodologies** → **Artificial Intelligence**.

KEYWORDS

Human-AI collaboration, Human-AI teams, Autonomous agent

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1 INTRODUCTION

Automated systems enhance analysis and decision making in domains ranging from social media [27, 81] to military intelligence [20, 21, 56]. However, these systems often encounter challenges that influence their efficacy, such as insufficient labeled data [59] or an inability to capture nuanced expert knowledge [35]. These limitations can be addressed by leveraging active learning techniques [63]. Active learning (AL) is a form of semi-supervised machine learning that incorporates an analyst's knowledge into the algorithm's training. The system queries the analyst to provide knowledge about data points that the system cannot classify with high certainty. The system integrates this new knowledge into its model to improve its performance. AL is especially important for applications where copious amounts of data are readily available, but the majority of that data is unlabeled due to the time and expertise costs associated with manual labeling.

We investigate how the ways people interface with active learning systems may influence an analyst's perceptions of system capabilities and their downstream interactions with the system. AL

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approaches often focus on choosing candidate data points that are likely to optimize algorithm performance, such as maximizing information gain [59]. However, such approaches focus exclusively on best strategies for the system and do not account for potential effects on the analyst and their confidence in the system. For example, presenting analysts with only low confidence examples may bias analysts' mental model of the system's capabilities, priming analysts to mistrust the system [3, 74]. Visualizations may help analysts better understand classifications by increasing model transparency [80], which may, in turn, influence the analyst's mental model of the system [83]. The reliance and trust exhibited by analysts is significantly influenced by their perceptions of the system's capabilities [23]. If the model performs well, analysts may place too much trust in its performance. Conversely, when the model exhibits suboptimal performance, analysts may develop a sense of mistrust towards the system. As a result, the model's performance on queried instances may directly affect trust in an AL system.

We conducted an empirical study to measure how query policies—the queried instances, classification confidence, and corresponding order in which systems query the analyst—and the methods used to communicate classifier confidence influence analyst trust in an automated image classifier's decisions. Trust in classification is critical in active learning applications as the analyst's interaction with the system does not stop at training [10, 68, 83]. Analysts are increasingly required to make decisions with data processed by autonomous tools, yet the analyst's interactions with a system can have unintended priming effects that may lead to mistrust or over-reliance [29]. However, conveying intuitive understandings of complex models and automation processes is challenging, and many basic questions surrounding cognitive bias and active learning remain unanswered. Query policies and classification visualizations are critical for mental model development and, as a result, may help calibrate trust in autonomous classification.

We explore query policies and their impact on analyst trust in a classification system post-training through an interactive image labeling task. Our experiment incorporates confidence-based query policies and visualizations sampled from existing practices in order to emulate a real-world interactive labeling environment, such as those used in medical image labeling [9] and satellite monitoring [37]. Our query policies explore a combination of high- and low-confidence examples and different ordering schemes to understand priming effects in AL applications. We couple these policies with six visualizations that provide a range of transparency into classification (see Figure 1). We found that while query policy and visualization type had little effect on objective indications of trust, they did impact people's confidence in the system's classification. We also found that more complex explanations may have negative impacts on classification performance. These results suggest an interplay between classifier explanations and querying strategies that may complicate existing guidelines for trust calibration in interface design.

Contributions: We measure how query policies and uncertainty visualizations of classification confidence influence analyst trust in autonomous systems using active learning. Our findings reveal the interdependence between these factors alongside several insights into how interface design could calibrate an analyst's level of trust

towards automation and labeling performance for people using AL systems to support classification tasks.

2 RELATED WORK

We review current studies at the intersection of visual analytics, HCI, and machine learning, with a focus on elucidating the role that visual representations play in human-centered machine learning. We then survey recent research assessing the level of trust analysts place in AI-assisted classification decisions in human-AI collaboration.

2.1 Visualization in Machine Learning

Visualization systems help people better understand the intricate processes inherent to ML algorithms [38, 62, 73]. Visualizations can both enhance analysts' comprehension of algorithm training dynamics and augment the interpretability of their outcomes [36, 39, 61]. While often employed for more explainable algorithms like SVMs [42], as deep learning has been adopted across diverse domains, visual analytics approaches have been developed to provide transparency into the functions of individual layers and parameters within neural network architectures [14, 66, 75]. Analysts can use these approaches to refine models in a range of applications [33, 45], concurrently lowering the barriers to entry for both experts and non-experts engaging in ML experimentation [32].

Visualizations are often used in explainable AI (XAI) and interactive machine learning for making a model's decisions easy to understand [28]. These visualizations can help people build better mental models of an algorithm, leading to more efficient human-AI teaming in a range of classification tasks [4, 36, 83]. Visualization can support analysts in more effectively leveraging machine learning within interactive workflows. For example, visualizations can convey information about a classifier's internal state [66, 75], its classification performance [40, 67], and can aid in model debugging [47]. These studies help bridge a potential gap between human analysts and AI-assisted classification models.

Many interactive ML applications use active learning, which involves real-time engagement with analysts and continuous refinement of the ML model [18]. This engagement can elicit active input about specific instances [8] or infer input from interactions with a system [25, 35]. Active learning and other human-in-the-loop approaches (see [51, 57] for surveys) can enhance performance [19, 72] and fairness [52] in classification tasks. However, the design of interfaces for working with these approaches shapes the mental models people build about the underlying ML model, which may impact the analyst's willingness to work with and trust the model [83].

2.2 Trust in Active Learning Systems

As datasets increase in size and complexity, active learning offers a promising strategy to reduce the effort required for generating labeled datasets by querying analysts to provide relevant information for important subsets of data [2]. Prior work in active learning has predominantly focused on devising model-centric approaches aimed at optimizing information gain while minimizing the number of instances queried from a human oracle [26]. For example, these AL systems often ask analysts to label data points where the

classifier has the greatest uncertainty [34, 43] or to select candidate data points representative of clusters of similar instances [84].

AL systems can also integrate human-centric modeling considerations [10], focusing on how model explanations and visualizations enhance AL model development. Tam et al. [68] compared these two methodologies for classification model construction task and found significant benefits from human-centric approaches, particularly in scenarios involving sparse datasets. However, the ways systems query for analyst input may negatively impact decision making by perpetuating preexisting bias [41, 44] or by leading to improper mental models [23, 54]. Further, analysts' trust in a classifier can shift their perceptions of when and where classifier input helps in a given decision making task [48].

Visualization design practices may substantially influence trust in a classifier [15, 24, 54], calibrating the user's mental model and the ways they use an ML model in practice (see Vereschak et al. [71] for a survey). For instance, Bansal et al. found that having access to classification explanations tended to make people more inclined to accept the model's decisions, regardless of their correctness or of the analysts' own expertise [5]. However, evidence for the role of explanations and trust in human-AI collaboration is mixed [22]. For example, Zhang et al. [83] found that explanations can increase trust depending on their design, but increased trust may not be sufficient to increase classification performance. Increased trust may even prove detrimental to analysis by perpetuating human bias [44]. The role of specific visualization design factors in analyst trust is complex and still not well understood [76].

While past work emphasizes the role of explanations or knowledge elicitation, this work tends to focus on the impact of each factor in isolation. We conducted an empirical study to investigate the interdependencies of different query policies and visualization methods. We aim to better understand how these factors collectively influence analyst trust to further inform the design of human-centered interactive labeling systems.

3 MOTIVATION & HYPOTHESES

AL systems typically present people with queries about data the system is uncertain about to maximize information gain [53]. However, presenting this uncertain information may lead to poor mental models of system behavior that may be improved by visualizations summarizing classifier reasoning [5, 10]. Strategies like employing explanations can enhance trust in AL systems and improve overall performance [18, 28, 76]. While XAI approaches can help people build better mental models to foster human-AI collaboration, mental models that leave people feeling the system is incapable of performing a task or even a failure to build a rationale mental model can significantly impair perceptions of AI capabilities and be hard for systems to overcome [4, 23].

This study is in part motivated by our collaborations with remote sensing analysts. These imaging applications often require specialized knowledge to label image contents over large numbers of images. Analysts may wish to employ a human-in-the-loop pipeline to scale their expertise with large sets of satellite images. Coupling the scalability of machine learning with expert interpretation can increase analysis efficiency while mitigating risks from incorrect classifications, but analysts must have appropriate trust

in a system's performance before they are willing to employ it. However, specialized applications require specialized participant pools, potentially introducing familiarity biases and inhibiting rigorous explorations of a range of system designs. In order to explore these effects in a larger, more generalizable population and set of designs, we simulate our collaborator's processes using more accessible stimuli (dog breeds) to remove biases present when examining domain-specific images and study a wider range of design choices.

In this study, we address three research questions relevant to analyst trust in active learning systems:

- (1) Will the query policies used to gather information from an analyst affect the analyst's confidence in the system during autonomous classification?
- (2) How will visualizations of classifier confidence influence an analyst's trust in autonomous classification?
- (3) Will the transparency provided by different visualization techniques shape the influence of different query policies?

We evaluate these questions using a range of designs inspired by both existing practices in active learning (see §5.1) and visualization (see §5.2). Based on prior work, we hypothesize that:

H1: Query policies that primarily present low confidence instances will reduce analyst trust in the classifier, resulting in system undertrust. People's decision making is heavily influenced by their first impressions of a system [1, 23]. Such priming effects may shape how an analyst applies a classifier after their preliminary training. We hypothesize that if analysts are presented with largely low confidence instances during early training, they will be less confident in the system downstream, even if the classifier performs well on later tasks.

H2: Visualizations that provide greater transparency into a model's classification process may lead to greater trust and better classification performance. Visualizations for machine learning applications often provide global context as a means of explaining how the classifier arrived at a certain decision [60]. Past studies show that richer explanations tend to lead to higher trust and performance [76, 83]. We anticipate that visualizations that lend greater global transparency into a classification will improve performance by better providing greater context for integrating model classifications into decision making.

H3: Visualizations situating examples within the full dataset will mitigate the impact of query policy. Offering explanations at different levels of locality can significantly influence how likely an analyst is to trust a system [50]. We anticipate that global explanations will provide context for low-confidence queries, providing greater situational awareness for queried instances. As such, we vary our visualization's level of locality with respect to the instance being labeled and the global population of images to determine if this will significantly offset effects imposed by our querying policy.

We evaluate these hypotheses in an 5 (query policy) $\times$ 6 (visualization type) between-subjects experiment using objective and subjective measures of trust and performance in an image classification task. We first constructed a dataset of human classification confidence to establish a ground-truth confidence that aligned with participant classification expectations (e.g., the confidences reported by the system matched participants' likely confidence in their own abilities to classify a given image, §4). We used these

confidences to simulate a synthetic classifier aligned with anticipated human confidence in classification (see §5). This ground truth enabled us to better align the synthetic active learning classifier’s performance with that of our participants. If the synthetic classifier were to perform better than the participant, it could result in the participant lending too much trust to the system, or too little if the participant performs better [5].

Our formal study asked participants to actively “train” the simulated classifier based on a set of queried examples using a specific query policy and visualization type. We then gauged participant trust in the resulting algorithm using a series of objective (accuracy and percentage agreement) and subjective (self-reported perceptions of confidence and trust) metrics.

4 DATASET CONSTRUCTION

We first collected data about human confidence on a target image corpus to generate baseline confidence scores for use in the primary experiment. We used these baseline confidence measurements to simulate classification in our primary study as opposed to a machine learning classifier’s confidence in order to mirror participants’ certainty in the classification and to avoid classifier-dependent results. This mirroring improved the likelihood that confidence correlated with task difficulty to mitigate confounds arising from confidence variations. For example, if a classifier had low confidence on an image people could easily classify, participants’ confidence in the system may degrade as a result of the system’s confidence in that individual image rather than their overall mental model of the system.

We establish this ground-truth confidence dataset by measuring human accuracy in classifying two pairs of dog breeds from the Oxford-IIIT Pet Dataset [55] at different downsampling levels. We collected responses from 178 participants (average age of 37.7; 85 males, 91 females, 2 no replies) via Amazon Mechanical Turk. 38 participants were excluded for labeling all images with the same label or failing two of four attention checks (i.e., misclassifying images with no downsampling).

4.1 Stimuli

We selected 100 images from four different dog breeds drawn from the Oxford-IIIT Pet Dataset [55]—a 37 category pet dataset with roughly 200 images for each category (see Figure 2). We divided images into two sets of breed pairs: Basset Hound/Saint Bernard and Chihuahua/Miniature Pinscher. These pairs and their respective images were chosen heuristically such that they shared a sufficiently large subset of common features that may make them challenging to visually discriminate. We sampled 25 images from each of the four dog breeds. To provide additional variance in classification difficulty, four new images were generated from each of the 25 images via downsampling using maxpooling operations at four different down-sampling rates: 2, 4, 7, and 10. For example, for a down-sampling rate of 2, 2×2 pixel grids would be pooled into a 1×1 pixel grid for the next image, where the 1×1 pixel is the maximum pixel value from the 2×2 pixel grid of the original, full-resolution image. This procedure generated a dataset of 400 images (2 breed pairs $\times$ 2 breeds per pair $\times$ 25 images per breed pair $\times$ 4 downsampling rates).

4.2 Procedure

We asked participants to identify whether or not a dog was of a specified breed. We collected data using breed as a between-subjects factor and downsampling rate as a within-subjects factor.

Our data collection procedure consisted of four phases: (1) informed consent, (2) training, (3) labeling, and (4) demographics. We divided 50 non-downsampled images per breed pair into two sets: 25 for training and 25 for labeling. Training and labeling sets were selected at random and counterbalanced between participants to ensure an even distribution of measures.

Participants first provided informed consent in accordance with our IRB protocol. The training phase taught participants to distinguish between the target breeds. Participants were presented with an example image for each of the two dog breeds they would see (i.e., Basset Hound/Saint Bernard or Chihuahua/Miniature Pinscher) at the top of the screen and a single test image of either the tested breed (“Dog Breed A”) or its complement within the tested set (“Not Dog Breed A”). Actual dog breed names were not used to provide consistency across datasets and to avoid providing additional semantic cues. Participants labeled the test image by pressing either key F (Dog Breed A) or J (Not Dog Breed A). These keys align with natural pointer finger positioning on a QWERTY keyboard. Each participant saw up to 25 non-downsampled images as the test images during the training phase. Participants correctly labelled 5 images from each breed before moving onto Phase 3, indicating they had sufficiently learned to discriminate between the two breeds.

In Phase 3, we removed the breed examples and asked participants to label each test image as either “Dog Breed A” or “Not Dog Breed A” using the same keypress inputs as in the tutorial. Participants completed this identification task for each of the remaining 25 images at each of the four tested downsampling rates, presented in random order to mitigate potential transfer effects. Four full resolution images were inserted at trial numbers 20, 40, 60, and 80 to serve as attention checks. Participants labeled a total of 104 images. They then provided basic demographic information.

4.3 Confidence Analysis

We computed confidence scores for each breed as the proportion of times each breed was labeled as “Breed A” for each of the four target breeds. If the image was labeled as “Not Dog Breed A” in more than 50% of samples, we assigned it a confidence of $1 - \text{prop}(\text{Breed A})$ and labeled the image as “Not Dog Breed A” in our simulated classifier. These confidence measures, summarized in Figure 3, serve as the confidence scores for the simulated classifier used in the primary experiment (see §5). Participants had a mean confidence for images in the breed pair for Set A with 82% mean confidence ($\sigma = 15\%$) and Set B with 71% mean confidence ($\sigma = 14\%$). The performance distribution within these datasets allows us to draw from a broad dataset of low- and high-confidence images for each breed pair in order to apply different query policies to the training phases of our primary experiment. We use the statistical mean of each dataset to divide our corpus into low- and high-confidence scores. This division created 104 and 108 low-confidence images and 96 and 92 high-confidence images for Sets A and B respectively.

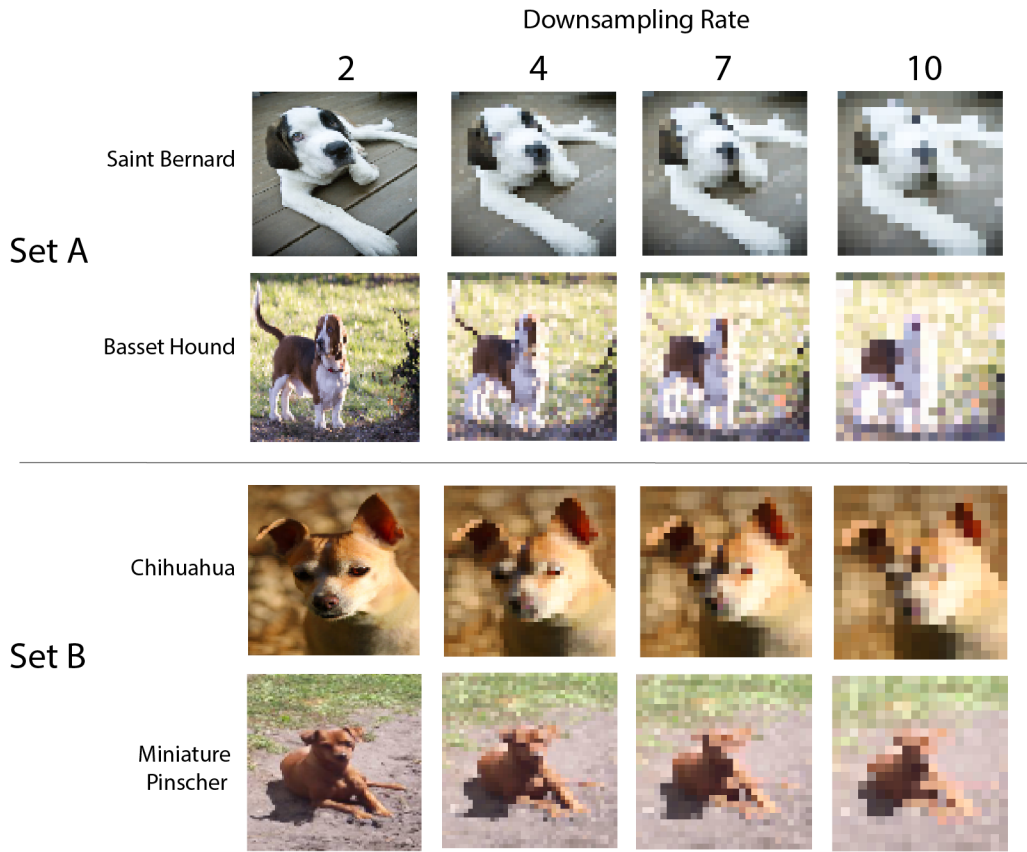


Figure 2: Representative images of dog breeds in our dataset. Participants were asked to classify between either Basset Hounds and Saint Bernards or Chihuahuas and Miniature Pinschers. To capture classifications across a range of confidence levels, participants completed these classifications for images at one of four downsampling rates.

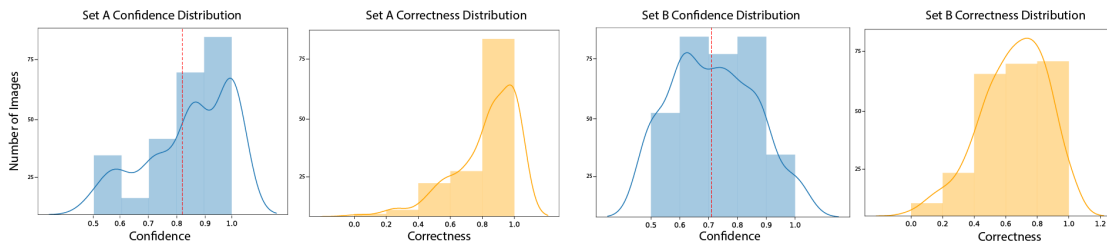


Figure 3: Distribution of confidence scores (shown in blue) and correctness scores (shown in orange) for each set of images. The dashed red line denotes the statistical mean of confidence values for each set, which we use to divide our corpus of images into low- and high-confidence sets. Note that confidence will always be above 50% as the confidence value matches the label provided by the majority of participants. Correctness will be below 50% if the majority of participants provide the incorrect label.

Our confidence measure is agnostic to whether the correct label was applied. We elected to use confidence rather than correctness as our primary metric to ensure our simulated classifier replicated the anticipated behaviors of the human analyst in our formal study. We also compute correctness scores as the proportion of times a breed was labeled correctly. Each image's correctness score is identical

to its confidence score if the majority of participants provided the correct label. We found 12 images for Set A (Basset Hound/Saint Bernard; 6% of all images in Set A) and 32 images for Set B (Chihuahua/Miniature Pinscher; 16% of all images in Set B) for which

the majority of participants provide the incorrect label. The respective confidence and correctness scores for each set of images are presented in Figure 3.

5 EXPERIMENTAL DESIGN

We used the results of our data collection study to simulate an active learning classifier leveraging different query policies and classification uncertainty visualizations in order to understand the role of query policy and visualization on analyst trust. We studied these effects using a 5 (query policy) $\times$ 6 (visualization type) mixed-factors experiment conducted on Amazon Mechanical Turk.

5.1 Query Policies

We tested five query policies, illustrated in Figure 4, to examine the potential impacts of query policy on analyst trust and performance. These policies reflect relevant past work in human learning [11–13] and cognition [69, 78] with an emphasis on patterns in high- and low-confidence query images (H1). The *random* policy, which randomly samples images from the full dataset, provides a baseline to understand the effects of other policies.

Query policies define the specific instances a system shows to an analyst. Only showing low confidence images, as in most AL sampling strategies [58], may cause analysts to build a negative mental model of the system’s capabilities by oversampling instances the system is uncertain about (i.e., selection bias). For example, an analyst may think the system is not confident in its classifications as they never see any evidence that the system has high confidence for the target classification. We anticipate that the inverse can also occur: only seeing high-confidence images may lead the analyst to believe the system is highly confident about the target task. We test *all low confidence (ALC)* and *all high confidence (AHC)* query policies that only query participants using images randomly sampled from the low or high confidence subsets respectively.

Anchoring bias—the tendency to rely more heavily on early information—may also influence trust in autonomous systems [54]. Studies from cognitive science indicate that the ordering in which people encounter instances may influence people’s mental model development [13]. Ordering effects may emerge in AL when systems query progressively more or less certain instances [64]. To evaluate these effects, we additionally test samples that query participants using increasing (*low-to-high confidence* or LtH) or decreasing (*high-to-low confidence* or HtL) confidence images. These policies use half low and half high confidence images and show the images to participants in increasing or decreasing order of confidence. Figure 4 presents examples for each respective query policy.

5.2 Visualization

We tested six different visualizations that span different levels of classification transparency (see Figure 1). Our *baseline* provides only a classification label with no information about classification uncertainty.

Two visualization conditions focus on only providing transparency into the specific instance confidence. The *numeric* visualization provides a raw confidence score, offering a basic level of transparency into the systems’ classification certainty. The *bar chart* visualizes this same confidence for the classification decision

alongside the confidence of the alternative classification using a simple bar chart.

Recent studies suggest that the type of transparency provided may also affect trust [28, 83]. We investigate this effect using three different means for contextualizing uncertainty with respect to the full dataset. The *1D dot plot* visualization contextualizes a given instance’s confidence in the distribution for the full dataset using a 1D dot plot, where points classified as “Dog Breed A” are shown in blue, those as “Not Dog Breed A” are shown in red, and the target instance is visualized as a large mark. The *t-SNE embedding plot* uses a common approach in modern visualization tools [7, 70]. This plot visualizes the set of images as a scatterplot in an embedded feature space, with coordinates representing images determined by image similarity values extracted in TensorFlow using ResNet [31]. In the t-SNE plot, images classified as “Dog Breed A” are shown in blue, “Not Dog Breed A” are shown in red, and the target instance is visualized as a large mark. The *exemplar plot* shows selected examples at different confidence levels (separated into quartiles) and the respective quartile for the current instance. This plot gives participants examples of each breed at different confidence thresholds to align their error boundaries with specific instances provided by the system [4] and potentially enhance trust through image-based explanations [76]. The exemplar plot uses images sampled at random from each confidence quartile arrayed along a horizontal axis from left (“Not Dog Breed A”) to right (“Dog Breed A”). Images are outlined using each breed’s corresponding color (blue or red).

These visualization types test a spectrum of classification transparencies, including local and global explanations, using designs that reflect past work. Local and global explanations are prevalent in mainstream explainable AI systems [49, 60]. Local explanations focus exclusively on information about the target instance (e.g., absolute classification confidence for that instance), allowing analysts to focus exclusively on the instance at hand. We use the numeric representation of certainty and bar chart visualization conditions (Figure 1b and 1c) to give context regarding a specific instance, emulating systems which provide context only at the local level [65]. Alternatively, global explanations situate an instance within the context of the larger dataset. The 1D dot plot and t-SNE embedding plot provide a global context for the current instance relative to the full population of instances, similar to LabelInspect [46]. The exemplar plot provides a hybrid representation, providing both local context by visually examining an instance and comparing it to similar instances of the same certainty and global context through visual comparison to instances of different certainty levels [30], similar to OoDAnalyzer [17].

5.3 Stimuli

We used the downsampled dog breed images from our collected datasets as our experimental stimuli (see §4). To test different query policies, we divided these stimuli into high and low confidence images. *High confidence* stimuli had confidence above their respective set mean, and *low confidence* fell below the statistical mean (see §4.3). We implemented the query policies and visualizations by sampling images within each breed based on these thresholds.

5.4 Experimental Design

We conducted a two-stage 5 (query policy) $\times 6$ (visualization type) full factorial between-participants experiment, with query policy and visualization type as our independent variables. We collected our dependent measures over a set of 15 images with a uniform distribution of high and low confidence. We adjusted the image labels provided by the simulated classifier by randomly sampling from a binomial distribution weighted according to the confidence score of the image (e.g., if the image confidence is 75%, the system will provide the correct label in 75% of instances).

Our dependent objective measures consisted of label disagreement—the number of times an analyst flipped the classification of a system-applied label—and accuracy—the proportion of provided labels that aligned with the correct image label. Prior work suggests that percentage agreement is a meaningful dependent measure of trust [77]: the more an analyst changes a label, especially when levels of self-certainty are high, the less the analyst is willing to rely upon the system. We collected two subjective measures per image—the participant’s confidence in their response and corresponding confidence in the system’s response—using a five-point Likert scale. We also collected data about the participant’s perceptions of the system using a post-study questionnaire. We analyzed our objective measures and per-question subjective measures using a two-factor (visualization method, query policy) full factorial ANOVA. We then used Tukey’s Honest Significant Difference test (HSD) with Bonferroni correction for all post-hoc analysis ($\alpha = .05$).

5.5 Experimental Procedure

The study consisted of five phases: (1) informed consent, (2) tutorial, (3) training, (4) testing, and (5) survey and demographics. Figure 5 summarizes the experimental process.

In Phase 1, participants provided informed consent and proceeded to the tutorial. In the tutorial, participants were presented with an example image for each of the two dog breeds at the top of the screen, a set of instructions describing the task, and a single image of either “Dog Breed A” or “Not Dog Breed A” with an accompanying visualization. As in the dataset construction procedure (§4), actual dog breed names were not used. A single image was presented in the middle of the screen that the participant must label using the same key press inputs as in the dataset construction procedure (keys F and J for Dog Breed A and Not Dog Breed A respectively). The images in the tutorial segment come from the set of 50 non-downsampled images per breed pair generated for the tutorial set, as described in §4.1. The participant continued labeling images until they correctly labeled 5 images from each breed, indicating they could discriminate between the two breeds. After completing the tutorial, participants received a brief explanation of the accompanying visualization before proceeding to Phase 3.

In Phase 3, participants were instructed that they would be training a classifier to better discriminate between the two dog breeds the participant was trained on, simulating an active labeling protocol. Each participant was presented 18 images sequentially, along with the system’s binary classification of the image—“Dog Breed A” or “Not Dog Breed A” as determined during dataset construction §4—and an accompanying visualization. We elected to use 18 training images based on piloting to maximize the number of

training images while maintaining engagement. The 18 training images are selected without replacement according to the query policy sampling described in §5.1 and §5.3. Each participant was randomly assigned one query policy and one visualization type, with combinations counterbalanced across participants.

Once the participant finished training the classifier on the 18 images, they proceeded to a holding page that instructed them that the classifier was updating based on their feedback. After the system trained, which took approximately 10 seconds with simulated progress indicated using a horizontal progress bar, the participant was instructed they would now evaluate the system’s performance.

In Phase 4, participants were again sequentially presented with 15 images to label. The 15 images corresponded to the test set images described in §5.4, with images randomly sampled and presented in a random order. Unlike in the previous segment, the participant was not provided with an accompanying visualization, enabling us to focus on trust constructed during active labeling. Showing classification confidence post-labeling integrates confidence into the testing phase and could interfere with primed perceptions. To ensure task engagement, image labels were selected by comparing image confidence to a random number and flipping the applied label if the number was larger than the confidence threshold. This strategy ensured disagreement correlated with image difficulty (i.e., overreliance would correlate with lower accuracy). For each image, participants labeled instances using a radio button and completed two five-point Likert questions pertaining to their confidence in their own decision (“How confident are you in your decision?”) and their confidence in the classifier’s decision (“How confident are you in the system’s decision?”). We randomly inserted four full-resolution images as engagement checks.

After completing the system evaluation stage, the participant proceeded to Phase 5, where they completed a brief post-survey questionnaire and demographics survey.

5.6 Participants

For the second study, we recruited 535 participants with a mean age of 39.72 ($\sigma = 11.19$) and gender distribution of 291 male, 238 female, 3 non-binary, and 4 no-response via Amazon Mechanical Turk. We excluded 72 participants for failing at least two of the three attention checks or labeling all images with the same label.

6 RESULTS

We analyzed our results from both the training and evaluation phases of the study. Data from the training phase confirms task engagement and comprehension. Data from the evaluation phase provides insight into classifier application post-training.

6.1 Training Phase

During the training phase, the proportion of disagreements—number of times the analyst flipped the label—aligned with expected behaviors based on the image distributions. We found a significant effect of query policy on disagreement ($F(5, 463) = 46.9974, p < .0001$) and on correctness ($F(5, 463) = 18.8507, p < .0001$). Participants disagreed with the baseline labels in the ALC condition ($\mu = .38, \sigma = .0114$) significantly more than in the AHC condition ($\mu = .161,$

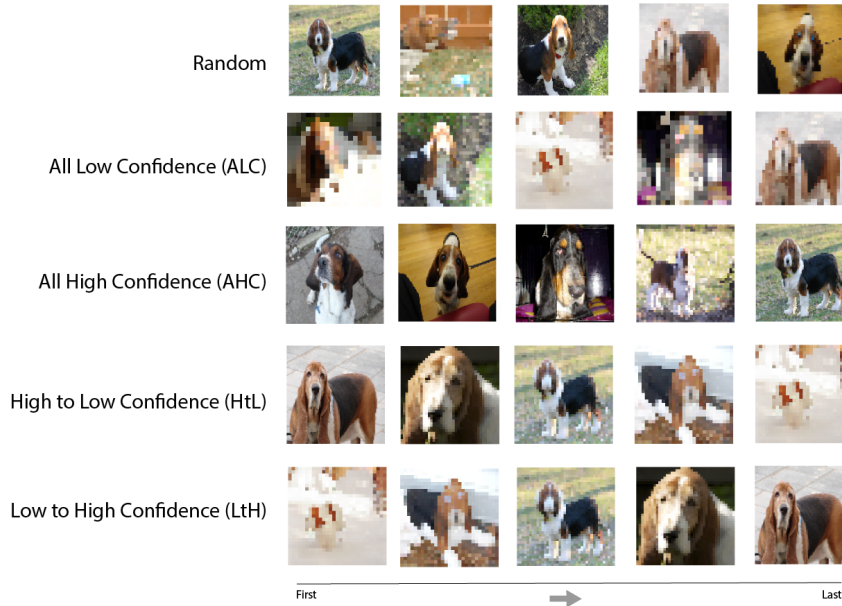


Figure 4: Examples of the five query policies used during the training phase of the primary study. Participants would see these instances sequentially, starting with the leftmost image. Note that while only five images per policy appear in the figure, 18 instances were shown during the training phase of the study.

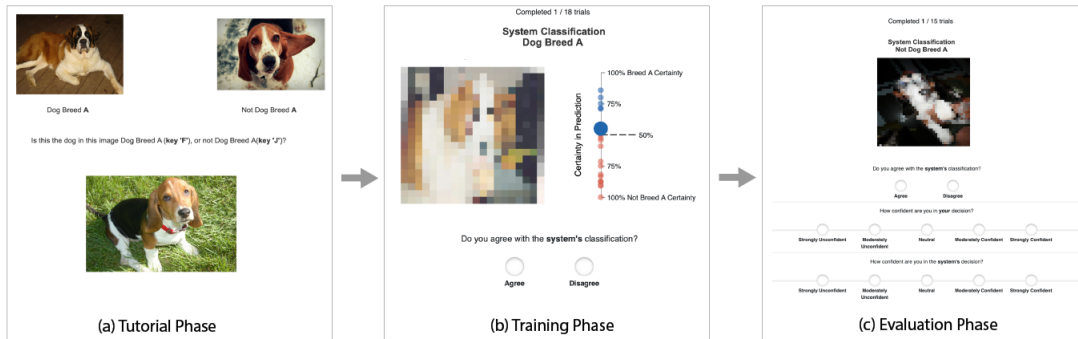


Figure 5: Example interfaces for Phases 2–4 of the primary study. (a) Tutorial phase: participants must correctly label five instances of each dog breed to proceed. (b) Training phase: participants must either agree or disagree with 18 per-image classifications from the system, presented according to the query policy and accompanied with an uncertainty visualization. (c) Evaluation phase: participants respond to 15 randomly drawn instances and self-evaluate their level of confidence in the system’s decision and their own decision.

$\sigma = .0085$). Disagreements for the HtL condition ($\mu = .285$, $\sigma = .0123$) and LtH condition ($\mu = .253$, $\sigma = .011$) were close to our random-baseline ($\mu = .274$, $\sigma = .012$). These results indicate that participants’ willingness to disagree with the classifier in training correlated with the number of low confidence images shown.

6.2 Evaluation Phase

We collected both objective and subjective measures per question in the evaluation phase. We additionally collected subjective data regarding overall model perceptions in a post-study questionnaire.

6.3 Objective Evaluation

We assessed participants’ behavior through three response variables: the rate of agreement with the classifier (i.e., how frequently the participant changes the system-provided label), the accuracy rate of the participant’s responses (i.e., how often a provided label matches the ground truth label of the image), and the average response time.

We did not find a significant effect of query policy or visualization type on label-change frequencies. All label-change frequencies were close to our random baseline ($\mu = .329$, $\sigma = .0470$). The frequency

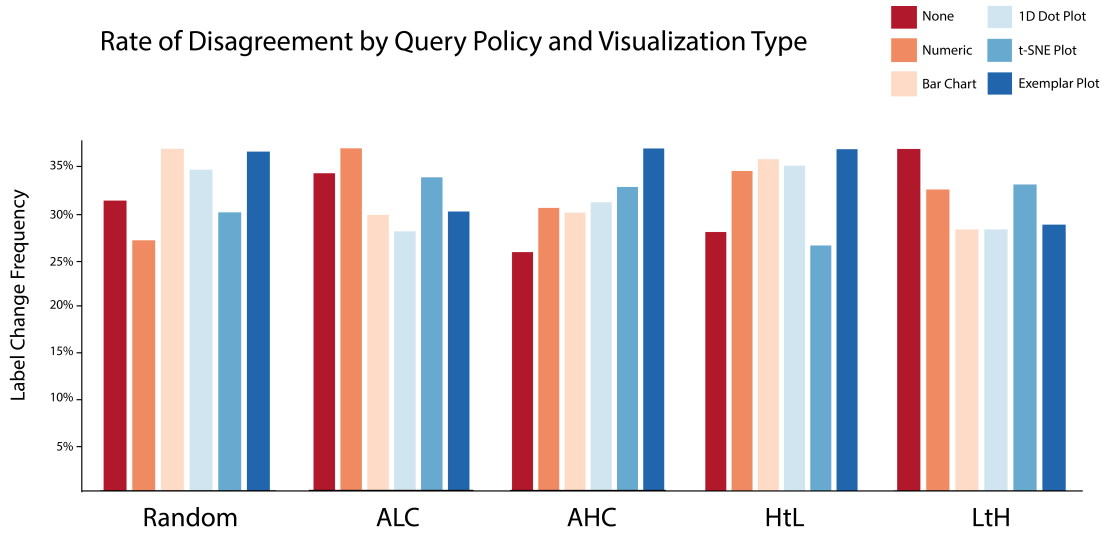


Figure 6: Rate of disagreement by query policy and visualization type during the evaluation phase of the primary study.

of disagreement for each policy and visualization type are shown in Figure 6.

We found a significant effect of visualization type on correctness during the evaluation phase ($F(5, 463) = 3.7549, p < .005$). Participants shown a numeric representation of classifier certainty during training achieved the highest correctness rate during the evaluation phase ($\mu = .729, \sigma = .044$), with the bar chart being second highest ($\mu = .708, \sigma = .045$). Both of these conditions led to significantly higher classification accuracy than the exemplar plot ($\mu = .646, \sigma = .047$). Participants with no accompanying visualization in the training phase had the third-highest level of correctness ($\mu = .704, \sigma = .405$) followed by the 1D dot plot ($\mu = .690, \sigma = .0463$) and the t-SNE Plot ($\mu = .676, \sigma = .046$).

While we found no significant main effect of query policy on correctness, the random-baseline policy paired with a numeric representation of confidence ($\mu = .781, \sigma = .041$) yielded significantly higher rates of correctness than AHC paired with an exemplar plot ($\mu = .590, \sigma = .049$).

We observed a significant effect of query policy on response times ($F(4, 463) = 2.199, p < .01$). Response times were lowest in our LtH policy ($\mu = 9.14s, \sigma = 1.41s$) and highest in our random-baseline policy ($\mu = 10.73s, \sigma = 1.61s$). We also found a significant interaction effect of query policy and visualization type on response times ($F(29, 463) = 2.2938, p < .005$). The LtH policies paired with a numeric representation of confidence ($\mu = 7.29s, \sigma = .467s$) or a 1D dot plot ($\mu = 8.17s, \sigma = .375s$) yielded significantly faster response times than our random-baseline paired with a t-SNE plot ($\mu = 13.05s, \sigma = 3.09s$) or an exemplar plot ($\mu = 12.19s, \sigma = 1.769s$) as well as our AHC policy paired with a numeric representation of confidence ($\mu = 12.48s, \sigma = 1.360s$). The magnitude of these differences suggests that people spent significantly more time actively reasoning over the model suggestions in conditions with longer response times.

6.4 Subjective Evaluation

We observed a significant effect of query policy on participants' self-reported confidence in the system's decisions ($F(4, 463) = 2.8226, p < .05$). Figure 7 shows the response distribution of participants' level of confidence in the system. Participants in the ALC condition were most confident in the system ($\mu = 3.815, \sigma = 1.091$), while participants were least confident in the system for AHC query policies ($\mu = 3.663, \sigma = 1.202$). The ALC policy was also significantly more confident than the random-baseline condition ($\mu = 3.669, \sigma = 1.177$). We did not find a significant main effect of visualization type.

We also found an interaction effect of query policy and visualization type on participants' level of confidence in the system ($F(29, 463) = 2.8491, p < .0001$). HtL queries paired with an exemplar plot yielded the lowest self-reported confidence, whereas the highest was seen in our HtL policy with no accompanying visualization. We found that HtL paired with a 1D dot plot ($\mu = 3.6719, \sigma = 1.0981$) or an exemplar plot ($\mu = 3.5179, \sigma = 1.1580$), ALC paired with no visualization ($\mu = 3.5227, \sigma = 1.1152$) or an exemplar plot ($\mu = 3.7024, \sigma = 1.0240$), AHC paired with a t-SNE plot ($\mu = 3.5417, \sigma = 1.0826$), and LtH paired with a 1D dot plot ($\mu = 3.5588, \sigma = 1.2678$) yielded significantly lower levels of system confidence as compared to all other conditions.

In contrast to perceptions of model confidence, we did not observe any statistically significant effect of query policy or visualization type on responses regarding participant's self-reported confidence in their own decision.

In our post-study questionnaire, we found that the ALC, HtL and LtH policies all had significantly higher responses to "I am confident in the system's decisions" compared to our random baseline ($F(4, 463) = 4.2833, p < .005$). These conditions had mean responses of $\mu = 3.837$ ($\sigma = .730$), $\mu = 3.713$ ($\sigma = .798$), and $\mu = 3.879$ ($\sigma = .728$) for ALC, HtL and LtH respectively. Similarly, we observed a significant effect of query policy on responses for

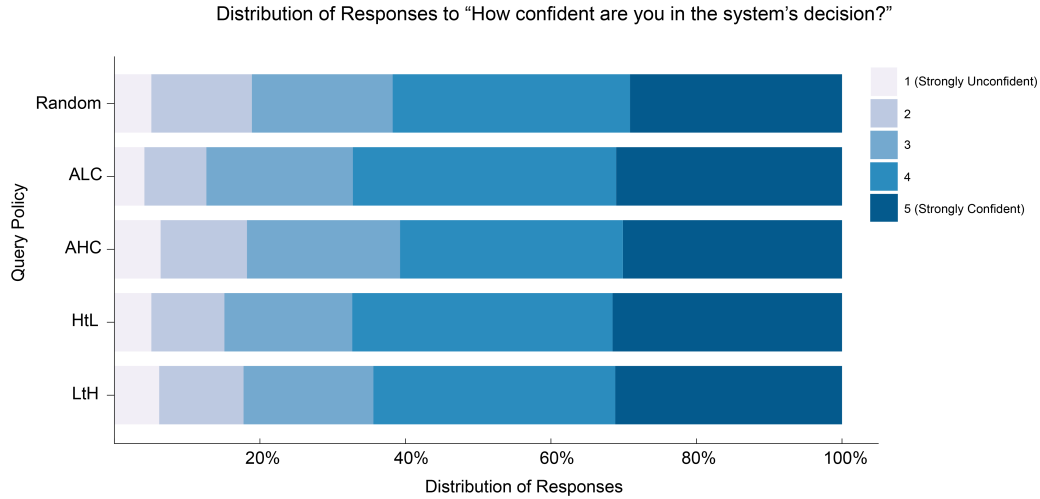


Figure 7: Distribution of responses for the per-instance subjective scale indicating participant’s confidence in the system, broken down by query policy. The All Low Confidence (ALC) policy yielded significantly more confident responses than the All High Confidence (AHC) policy and our random-baseline policy.

“The system’s output is trustworthy” ($F(4, 463) = 3.8683, p < .005$) where the HtL ($\mu = 3.819, \sigma = .855$) and LtH ($\mu = 3.824, \sigma = .783$) policies result in a significantly higher trustworthiness than our random baseline ($\mu = 3.514, \sigma = .905$). Query policy also had a significant effect on responses to “I feel like I understood the system’s confidence” ($F(4, 463) = 2.8802, p < .05$). We found that the responses to the random baseline policy ($\mu = 3.598, \sigma = 1.017$) were significantly lower than that of the HtL policy ($\mu = 3.904, \sigma = .881$). The distribution of responses to these prompts are shown in Figure 8.

We found marginal effects of visualization type on perceived system trustworthiness ($F(5, 434) = 1.9681, p < .09$). Perceived trustworthiness was highest when t-SNE plots, exemplar plots, or numeric text were used than when no information was provided or when a bar chart was used.

We found a significant interaction effect ($F(20, 463) = 1.694, p < .05$) between query policy and visualization type in perceived confidence in the system’s decisions and a marginal effect ($F(20, 463) = 1.4683, p < .09$) in people’s perceived comprehension of system confidence. The ALC policy paired with a t-SNE plot ($\mu = 4.384, \sigma = .506$) and the HtL policy paired with numeric representation of certainty ($\mu = 3.812, \sigma = 1.16$) yielded significantly higher confidence in the system as compared to all other conditions, while our random-baseline policy paired with a bar chart yielded significantly lower responses than all other conditions ($\mu = 3.50, \sigma = .894$). Similarly, people felt that they understood the system’s confidence better when using HtL with text than using a random baseline with a bar chart. We found no significant differences across query policy or visualization type for the prompts “The system is too confident in its own decisions” and “I feel that I rely on the system when I am uncertain”.

7 DISCUSSION

We explore the interaction of explanation design and query policy through an active learning scenario for image classification. This study adds to a growing body of work investigating how system design influences human-AI teaming. While our experimental methodology is limited to a single dataset and the task of interactive image labeling, our approach provides preliminary insights into trust in AL systems in a task setting that minimizes dependence on prior expertise, model selection, and other potential confounding factors. Current active learning approaches typically rely on selecting instances which are optimal for the learning algorithm, yet potential effects of these choices on people are underexplored. We hypothesized that policies based on querying instances with a high-degree of uncertainty may significantly degrade analyst trust in the classifier and that explanations providing global insight into system classification may mitigate lost trust. However, our results suggested querying on lower confidence images may increase trust overall. We also found evidence that more structured query policies may benefit perceived trust and that richer explanations of system confidence may represent trade-offs in performance and user confidence.

7.1 Confidence Distributions Alone Do Not Predict Trust

Query policies offer a small number of concrete examples to ground people’s perceptions of a model. While we had anticipated that query policies sampling low confidence instances would give an impression of the system being less capable (H1), we found the opposite: people were substantially less confident in systems only querying high confidence examples and were more confidence in all low-confidence queries compared to a random confidence distribution.

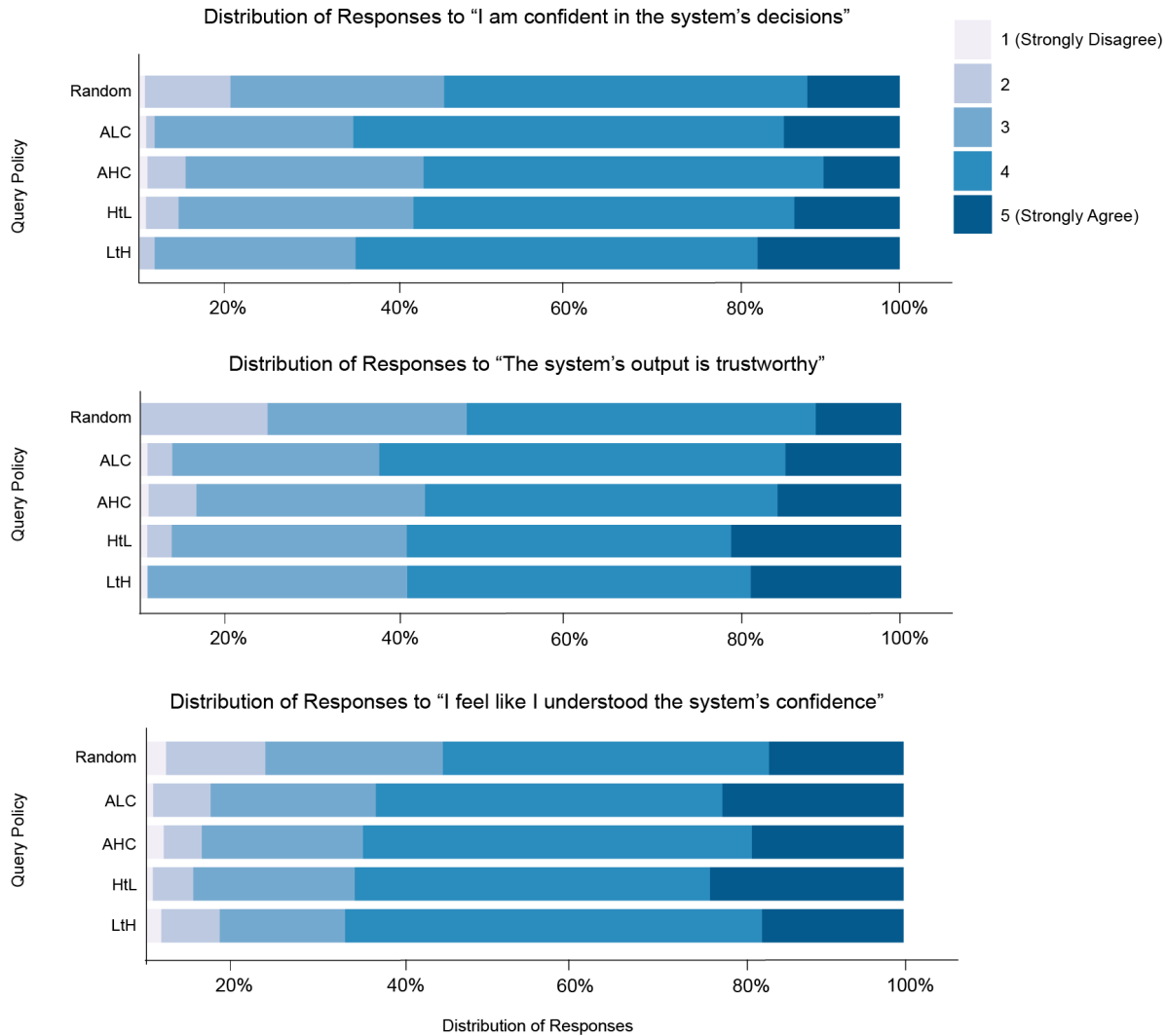


Figure 8: Distribution of responses to three of the five questions in the post-study questionnaire. We observed significant differences between policies in these three questions, primarily between the random-baseline policy and our LtH and HtL policies. People had significantly higher perceived confidence in systems trained with ALC, HtL, and LtH than the random baseline. They found systems trained with LtH and HtL significantly more trustworthy than the random baseline. They felt they best understood the system's confidence with HtL policies.

One possible explanation for this phenomena is that when participants witness the classifier seeking confirmation for easily identifiable high-confidence examples, they feel that the system requires assistance with tasks that should be straightforward [5]. However, our random, HtL, and LtH policies also included several high-confidence examples and performed comparably to the ALC condition across most measures. Alternatively, people may have been second guessing their own abilities relative to the systems' for the AHC condition. As we used human confidence scores as our simulated classifier, the confidences we see in the system aligned with the confidences that people felt. For the harder instances in

the evaluation phase, people in the AHC condition may not have had prior evidence of just how challenging the tested task could be and were unable to build up a mental model of how the system performed with challenging examples [23].

We found evidence that the order in which systems queried people may be a more significant factor in trust than the distribution of confidence scores. Specifically, LtH and HtL tended to generate faster classifications with greater perceived confidence and trustworthiness in the system. People also felt they better understood the confidences generated in the HtL condition. These ordering effects may indicate that people can more effectively construct a mental

model of the system’s confidence. People may see the systematic changes in queried examples as evidence of the system updating and adjusting its own level of confidence and “learning” from their responses. Alternatively, people may see the system as querying “intelligently” and anthropomorphize the system as a result. This hypothesis aligns well with past work that demonstrates the benefits of machine teaching paradigms on trust in active learning [28]: system behaviors perceived as evolving or intelligent may lead to more favorable impressions of the system overall.

7.2 Complex Explanations Do Not Always Benefit AL

While we had hypothesized that greater transparency into model classification would improve objective and subjective performance (H2), we found little effect of visualization type on our dependent measures. In contrast to prior work [83], visualizations providing more insight into the full dataset led to worse performance on the labeling task. This phenomena could indicate that participants are relying too heavily on the accompanying visualization rather than on the target image and do not sufficiently learn the classification task [39].

These more complex visualizations significantly degraded confidence in otherwise trustworthy policies but did not enhance confidence in low-trust policies, partially supporting H3. For example, while people were highly confident in the HtL policy with no visualization, adding an exemplar plot to the same policy led to one of the lowest overall confidence conditions. Exemplar plots tended to degrade confidence of high confidence policies and lead to more classification errors overall. We anticipate that these plots may have been especially prone to overreliance: people may have adjusted their model for the task based on comparing to the small subset of images in the visualization rather than their internal model of breed differences. Given the value of exemplars in past studies [76] and techniques [60], these effects should be studied further to better understand how the ways interfaces employ exemplars can (and cannot) support trustworthy classification. In real-world interactive labeling systems, where sampling strategies may be far more complex than our confidence-based model, visualizations may benefit from being intentionally designed to support the specific sampling methodology. For example, active learning is commonplace in scenarios where image recognition models must be fine-tuned on imagery which both humans and the learning algorithm struggle to classify and use attributes of the data to bootstrap sampling [6]. As such, more complex visualizations grounded in exemplar images that highlight data attributes guiding the sampling may be better suited to these labeling scenarios.

7.3 Limitations and Future Work

This work presents an experimental study measuring how an analyst’s level of trust in an automated image classification system may shift according to the ways the system interfaces with an analyst. Exploring over- or under-reliance is crucial to designing human-centric AI systems, as the effects of overreliance can have severe consequences in many domains. Our work provides several insights into what factors influence analyst trust in active learning

and interactive labeling systems; however, it also confirms that the interplay between these factors is complex [76].

Our study focuses on static, controlled interactions between the analyst and classifier to understand specific biases in human-AI interaction. Real-world systems may offer a range of additional interactions, such as letting people choose relevant data to label using visualizations of multiple instances [16] or supporting natural language dialogue between the analyst and the model [82]. The scope of our experimental design is limited to the task of interactive image labeling. Numerous domains employ active learning for improving model performance across a range of interactions. Future work should explore how these alternative interactions, data types, and the agency they impart may influence classification. This could also include more extensive testing methodology through longer-term interactions with automated systems rather than shorter interactions as tested in this work or variations introduced by using real-world classifiers whose performance may not align with human expectations.

We explore overreliance and automation bias through an experimental setup simulating an image classification system for commonly recognizable animals. We chose this setup as it does not require extensive domain-specific knowledge to perform accurately, allowing us to understand active learning agnostic to domain expertise and to use Mechanical Turk to collect data from a large and diverse subject pool. Active learning systems are commonly used in applications where domain-specific knowledge is needed, such as medical imaging [9]. Future work could explore more domain-specific applications of active learning to evaluate the role of expertise—both in machine learning and in domain sciences—and data—type, decision risk, and semantics—on trust and classification. Our study also modeled biases over a short time frame—18 training questions followed immediately by testing. Biases often accumulate over time, meaning that understanding the development, evolution, and persistence of these biases over time is important future work.

We specifically focus on confidence-based policy modeling. In real-world active learning systems, sampling strategies may select instances according to a variety of different policies and attributes of the domain, data, and task. Future work should explore the modeling of other sampling strategies, such as diversity-based sampling [79], to explore how a broader range of query policies influence analyst trust.

7.4 Conclusions

Active learning and human-in-the-loop machine learning systems are becoming increasingly commonplace. These systems, which rely on iterative feedback from analysts, can significantly enhance human-AI collaboration and classification performance in various applications. However, effectively using active learning strategies requires careful consideration of analyst trust, a complex factor that can impact classification performance.

Our study explores the interplay between query policies and visualizations of system certainty in shaping analyst trust and performance in classification. We found that the ways in which AL systems elicit input can significantly influence an analyst’s perception of system capabilities, potentially leading to over- or under-use

of system classifications. Providing analysts with different levels of transparency can significantly impact their ability to perform classification tasks effectively. These findings emphasize the need for a holistic approach to the design of interactive labeling systems that takes into account both the ways in which systems request feedback and communicate their decision making. We anticipate better understanding these trade-offs will lead to more robust and trustworthy active learning systems.

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